STANDARD 2

Stage 6

Networks (Std 2), N2 Network Concepts (Y12)

Basic Concepts

Teacher: Hailey Summer

Exam Equivalent Time: 69 minutes *(based on allocation of 1.5 minutes per mark)*

STD2 Exam Contribution History
N2 Network Concepts

Basic Concepts
Shortest Paths
Minimum Spanning Trees

***SmarterMaths analytics based on the average contribution in new syllabus exams (since 2019).**

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

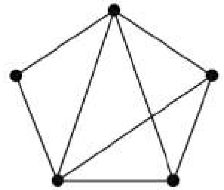
- *MS-N2 Network Concepts* has contributed 5.0% to new syllabus exams since its introduction in 2019.
- We have split this area into three categories for the purposes of analysis: *1-Basic Concepts (1.4%), 2-Shortest Paths (0.8%) and 3-Minimum Spanning Trees (2.8%).*
- This analysis will look at *Basic Concepts (1.4%).*

ANALYSIS - What to Expect and Common pitfalls

- **Draw network from table/matrix:** Completing a network graph from a table/matrix is a core competency which has been examined in longer answer questions in both 2022 and 2023.
- The database also covers the reverse process of completing a table related to a given network.
- **Basic Concepts:** Questions review basic concepts such as; odd and even vertices, tree and path definitions, connected network, weighted edge, etc..
- A disproportionate number of multiple choice questions are used in our database to allow teachers to target specific basic concepts.
- **Practical problems:** a number of practical problems requiring simple network design and representation are included. A review of *2020 Std2 9 MC* exposed a lack of understanding of "competition" networks and should be reviewed.

Questions

1. Networks, STD2 N2 2010 FUR1 2 MC



The number of edges in the graph above is

A. 5

B. 7

C. 8

D. 10

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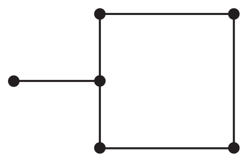
2. Networks, STD2 N2 2012 FUR1 1 MC



The sum of the degrees of all the vertices in the graph above is

A. 6

B. 9

C. 11

D. 12

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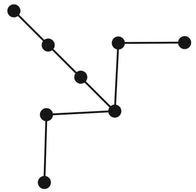
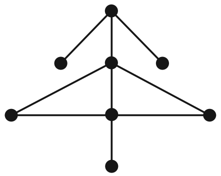
3. Networks, STD2 N2 2013 FUR1 1 MC

Which one of the following graphs is a tree?

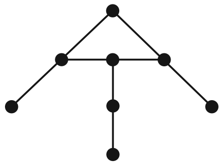
A.



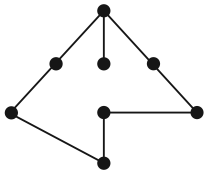
B.



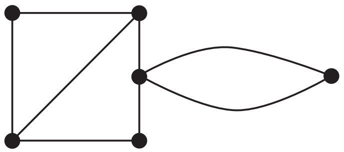
C.



D.



4. Networks, STD2 N2 2015 FUR1 1 MC



In the graph above, the number of vertices of odd degree is

A. 1

B. 2

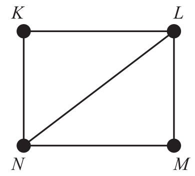
C. 3

D. 4

5. Networks, STD2 N2 2015 FUR1 5 MC

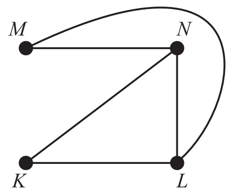
The graph below represents a friendship network. The vertices represent the four people in the friendship network: Kwan (K), Louise (L), Milly (M) and Narelle (N).

An edge represents the presence of a friendship between a pair of these people. For example, the edge connecting K and L shows that Kwan and Louise are friends.

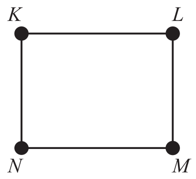


Which one of the following graphs does **not** contain the same information?

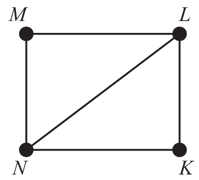
A.



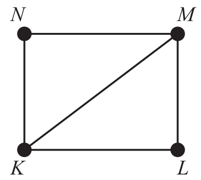
B.



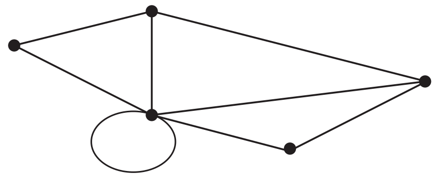
C.



D.



6. Networks, STD2 N2 SM-Bank 32 MC



The number of vertices with an odd degree in the network above is

A. 2

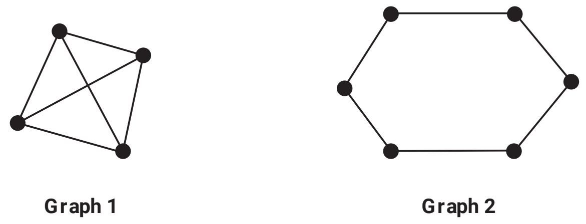
B. 3

C. 4

D. 5

7. Networks, STD2 N2 2017 FUR1 2 MC

Two graphs, labelled Graph 1 and Graph 2, are shown below.



The sum of the degrees of the vertices of Graph 1 is

A. two less than the sum of the degrees of the vertices of Graph 2.

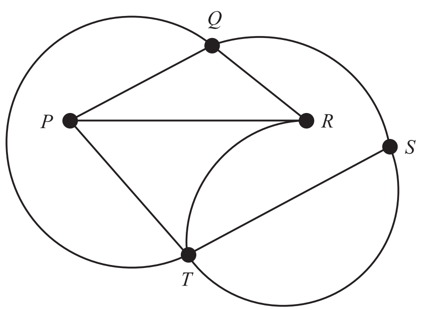
B. one less than the sum of the degrees of the vertices of Graph 2.

C. equal to the sum of the degrees of the vertices of Graph 2.

D. two more than the sum of the degrees of the vertices of Graph 2.

8. Networks, STD2 N2 2018 FUR1 4 MC

Consider the graph below.



Which one of the following is **not** a path for this graph?

A. $PRQTS$

B. $PRTSQ$

C. $PTQSR$

D. $PTRQS$

9. Networks, STD2 N2 SM-Bank 3 MC

A store manager is directly in charge of five department managers.

Each department manager is directly in charge of six sales people in their department.

This staffing structure could be represented graphically by

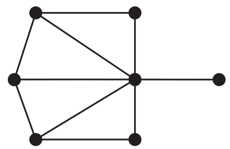
A. a tree.

B. a path.

C. a cycle.

D. a weighted graph.

10. Networks, STD2 N2 2011 FUR1 1 MC



In the network shown, the number of vertices of even degree is

A. 2

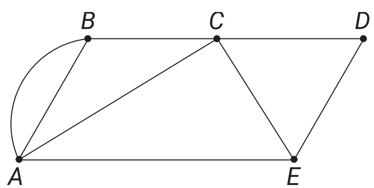
B. 3

C. 4

D. 5

11. Networks, STD2 N2 2021 HSC 2 MC

Consider the network diagram.



What is the sum of the degrees of all the vertices in this network?

A. 5

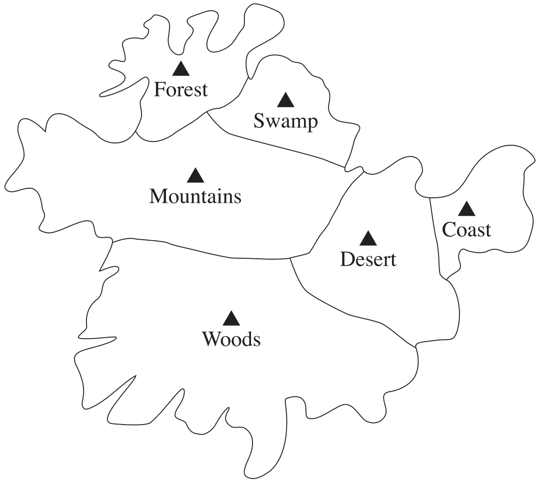
B. 8

C. 14

D. 16

12. Networks, STD2 N2 2024 HSC 4 MC

The map shows regions within a country.



A network diagram is to be drawn to represent this map. Vertices will be used to indicate each region and edges will be used to represent a border shared between two regions.

How many edges will there be in the network diagram?

A. 8

B. 7

C. 6

D. 5

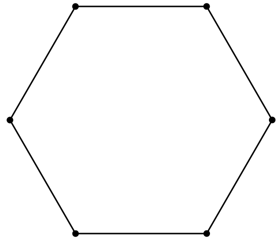
13. Networks, STD2 N2 2020 HSC 9 MC

Team $\boldsymbol{A}$ and Team $\boldsymbol{B}$ have entered a chess competition.

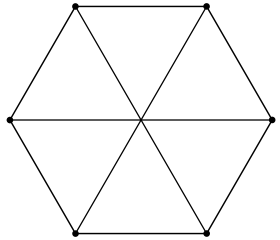
Team $\boldsymbol{A}$ and $\boldsymbol{B}$ have three members each. Each member of Team $\boldsymbol{A}$ must play each member of Team $\boldsymbol{B}$ once.

Which of the following network diagrams could represent the chess games to be played?

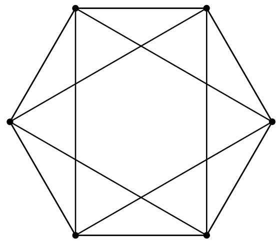
A.



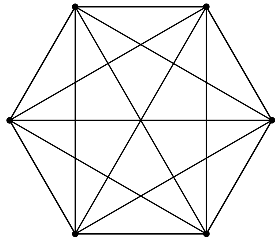
B.



C.



D.



14. Networks, STD2 N2 2009 FUR1 8 MC

An undirected connected graph has five vertices.

Three of these vertices are of even degree and two of these vertices are of odd degree.

One extra edge is added. It joins two of the existing vertices.

In the resulting graph, it is **not** possible to have five vertices that are

A. all of even degree.

B. all of equal degree.

C. one of even degree and four of odd degree.

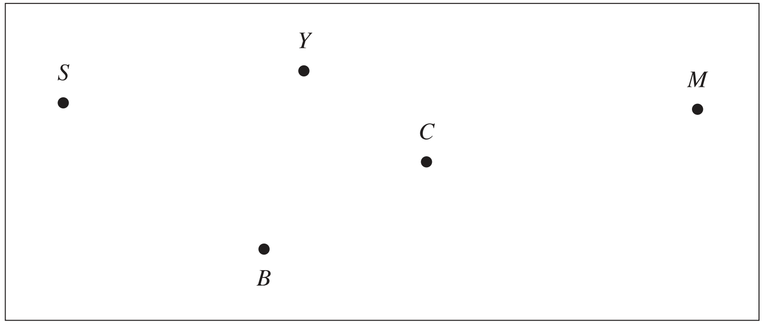
D. four of even degree and one of odd degree.

15. Networks, STD2 N2 2022 HSC 20

The table below shows the distances, in kilometres, between a number of towns.

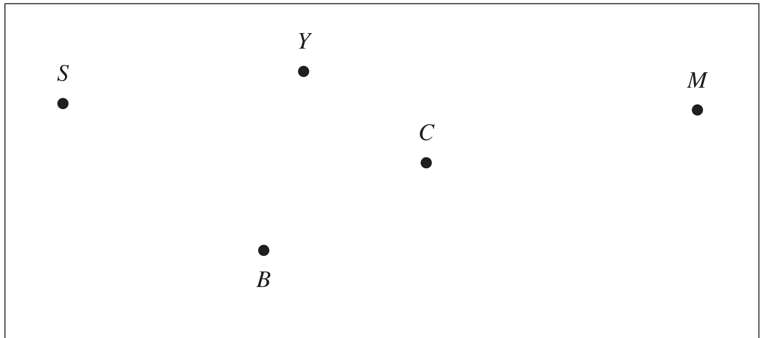
Towns	Snowtown (S)	Clairville (C)	Yuma (Y)	Bosten (B)	Morrella (M)
(S)	–	–	280	275	–
(C)	–	–	60	150	–
(Y)	280	60	–	–	530
(B)	275	150	–	–	790
(M)	–	–	530	790	–

a. Using the vertices given, draw a weighted network diagram to represent the information shown in the table. ***(2 marks)***



b. A tourist wishes to visit each town.

Draw the minimum spanning tree which will allow for this AND determine its length. ***(3 marks)***



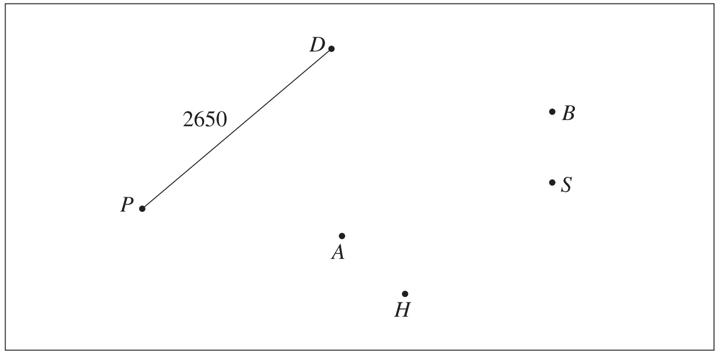
Length of minimum spanning tree = ...

16. Networks, STD2 N2 2023 HSC 17

The table shows some of the flight distances (rounded to the nearest 10 km) between various Australian cities.

City	Adelaide (A)	Brisbane (B)	Darwin (D)	Hobart (H)	Perth (P)	Sydney (S)
Adelaide		–	–	1170	2120	–
Brisbane	–		2850	–	–	750
Darwin	–	2850		–	2650	3150
Hobart	1170	–	–		–	1040
Perth	2120	–	2650	–		3270
Sydney	–	750	3150	1040	3270	

a. Use the information in the table to complete the network diagram where the edges are labelled with distances. ***(2 marks)***

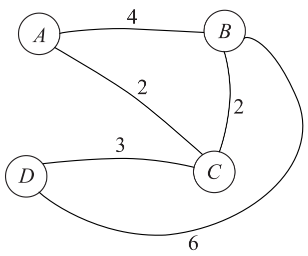


b. Mahsa wants to travel from Hobart to Darwin. She wants to change planes only once.

Using the network diagram, calculate how many kilometres she will travel by plane. ***(1 mark)***

17. Networks, STD2 N2 SM-Bank 20

A table is constructed to represent the network diagram below.



Complete the table. ***(2 marks)***

	A	B	C	D
A	0	4		-
B	4	0		
C			0	3
D		6		0

18. Networks, STD2 N2 SM-Bank 28

In central Queensland, there are four petrol stations A, B, C and D. The table shows the length, in kilometres, of roads connecting these petrol stations.

	A	B	C	D
A	–	170	–	150
B	170	–	160	90
C	–	160	–	120
D	150	90	120	–

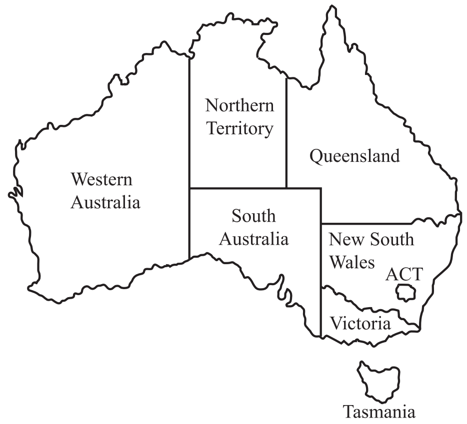
a. Construct a network diagram to represent the information in the table. ***(2 marks)***

b. A petrol tanker needs to refill each station. It starts at Station A and visits each station.

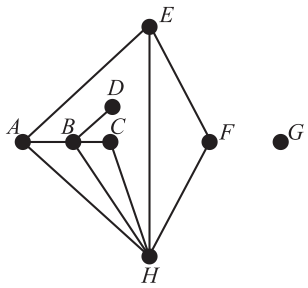
Calculate the shortest distance that can be travelled by the petrol tanker. In your answer, include the order the petrol stations are refilled. ***(2 marks)***

19. Networks, STD2 N2 SM-Bank 37

The map of Australia shows the six states, the Northern Territory and the Australian Capital Territory (ACT).



In the network diagram below, each of the vertices A to H represents one of the states or territories shown on the map of Australia. The edges represent a border shared between two states or between a state and a territory.



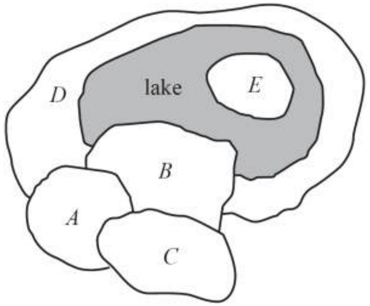
i. In the network diagram, what is the order of the vertex that represents the Australian Capital Territory (ACT)? ***(1 mark)***

ii. In the network diagram, Queensland is represented by which letter? Explain why. ***(2 marks)***

20. Networks, STD2 N2 2009 FUR2 1

The city of Robville is divided into five suburbs labelled as A to E on the map below.

A lake which is situated in the city is shaded on the map.



A table is constructed to represent the number of land borders between suburbs.

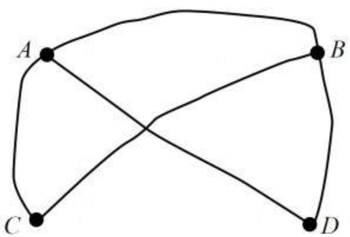
If there is no land border between two suburbs, the table records a '0'. If there is a single land border between two suburbs, the table records a '1', and if there are two separate land borders between the same two suburbs, the table records a '2'.

$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

a. Explain why all values in the final row and final column of the table are zero. ***(1 mark)***

In the network diagram below, vertices represent suburbs and edges represent land borders between suburbs.

The diagram has been started but is not finished.



b. The network diagram is missing one edge and one vertex.

On the diagram

i. draw the missing edge ***(1 mark)***

ii. draw and label the missing vertex. ***(1 mark)***

21. Networks, STD1 N1 2019 HSC 17

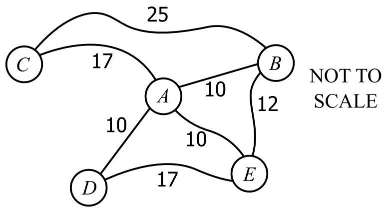
A regional airline operates flights in Queensland. Flight times between connected towns are shown in the table.

	Cairns	*Kowanyama*	*Mt Isa*	*Pormpuraaw*	*Townsville*
Cairns	–	1 h 50 min	2 h 5 min	–	55 min
Kowanyama	1 h 50 min	–	–	20 min	–
Mt Isa	2 h 5 min	–	–	–	1 h 40 min
Pormpuraaw	–	20 min	–	–	–
Townsville	55 min	–	1 h 40 min	–	–

Draw a network diagram to show how the towns are connected, with weights on the edges showing the flight times. ***(2 marks)***

22. Networks, STD2 N2 SM-Bank 11

A network of roads between towns shows the travelling times in minutes between towns that are directly connected.



Complete the shaded cells in the following table so that it represents the information in this network. ***(2 marks)***

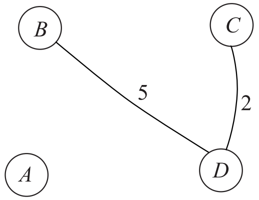
	A	*B*	*C*	*D*	*E*
A	0	10		10	10
B		0			
C	17		0	—	
D		—	—	0	17
E	10				0

23. Networks, STD2 N2 SM-Bank 21

The table below represents a network.

	A	B	C	D
A	-	0	7	4
B	0	-	3	5
C	7	3	-	2
D	4	5	2	-

Complete the network diagram below to accurately reflect the information in the above table. ***(2 marks)***

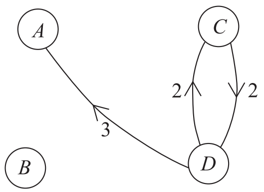


24. Networks, STD2 N2 SM-Bank 22

The table below represents a directed network.

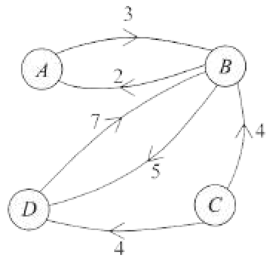
		To			
		A	B	C	D
From	A	0	4	-	4
	B	-	0	-	3
	C	-	6	0	2
	D	3	5	2	0

Complete the network diagram below to accurately reflect the network described in the above table. ***(2 marks)***



25. Networks, STD2 N2 SM-Bank 23

A directed network diagram is pictured below.



The information in the network diagram is used to complete the network table below, with a "0" used to signify that no connection exists. Complete the table. ***(2 marks)***

		To			
		A	B	C	D
From	A	-	3		0
	B	2	-		
	C			-	4
	D				-

26. Networks, STD2 N2 SM-Bank 40

The map below shows seven countries within Central America.

Draw a network diagram of the map where seven vertices represent each of the countries on the map and edges represent a border shared between two countries. ***(2 marks)***

Worked Solutions

1. Networks, STD2 N2 2010 FUR1 2 MC

Edges are represented by lines between vertices.
$\Rightarrow\ C$

2. Networks, STD2 N2 2012 FUR1 1 MC

Total Degrees
$= 1+3+2+2+2+2$
$= 12$
$\Rightarrow D$

3. Networks, STD2 N2 2013 FUR1 1 MC

A tree cannot contain a cycle.
$\Rightarrow\ A$

4. Networks, STD2 N2 2015 FUR1 1 MC

$\Rightarrow B$

5. Networks, STD2 N2 2015 FUR1 5 MC

Option D has Kwan and Milly as friends which is not correct.
$\Rightarrow D$

6. Networks, STD2 N2 SM-Bank 32 MC

$\Rightarrow A$
(Note a loop creates 2 extra degrees to a vertex.)

7. Networks, STD2 N2 2017 FUR1 2 MC

Graph 1
$\sum$ degrees $= 3+3+3+3 = 12$
Graph 2
$\sum$ degrees $= 2+2+2+2+2+2 = 12$
$\Rightarrow C$

8. Networks, STD2 N2 2018 FUR1 4 MC

By trial and error:
Consider option C,
$PTQSR$ is not a path because S to R
must go through another vertex.
$\Rightarrow C$

9. Networks, STD2 N2 SM-Bank 3 MC

$\Rightarrow\ A$

10. Networks, STD2 N2 2011 FUR1 1 MC

Vertices with even degrees: 2, 2, 6
$\Rightarrow\ B$

11. Networks, STD2 N2 2021 HSC 2 MC

Working from A to E:

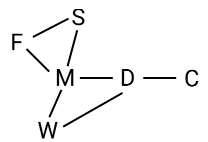
Sum of degrees $= 4+3+4+2+3$
$= 16$

$\Rightarrow D$

COMMENT: This simple question caused problems for many with mean mark just 58%.

12. Networks, STD2 N2 2024 HSC 4 MC

Network diagram:



Network has 7 edges.
$\Rightarrow B$

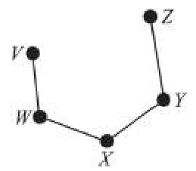
13. Networks, STD2 N2 2020 HSC 9 MC

Vertices $=$ players
Edges $=$ games between 2 players
Since each player plays once against the three players
in the other team, each vertex must be degree 3.
$\Rightarrow\ B$

♦ *Mean mark 38%.*

14. Networks, STD2 N2 2009 FUR1 8 MC

Consider an example of the graph described (below):

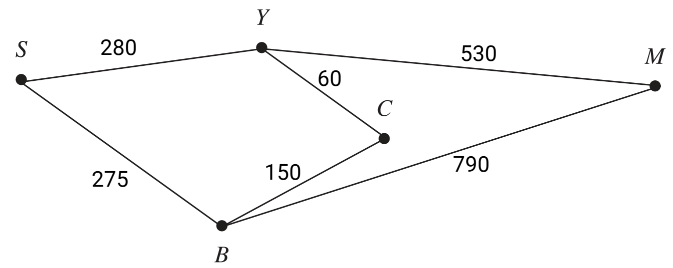


♦♦♦ *Mean mark 25%.*

A is possible – join V and Z
B is possible – join V and Z
C is possible – join W and Y
D is NOT possible
$\Rightarrow\ D$

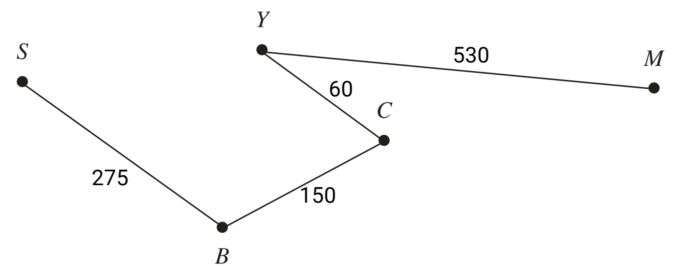
15. Networks, STD2 N2 2022 HSC 20

a.



b. Using Prim's algorithm (starting at Y):

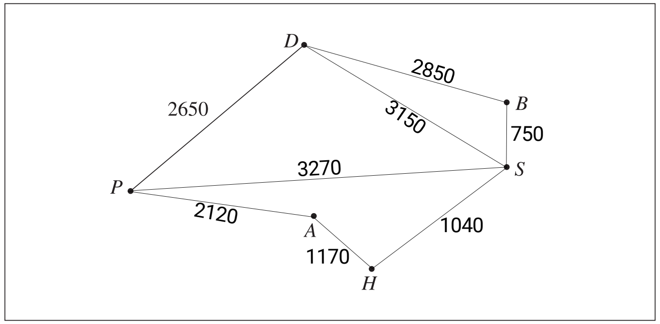
1st edge: YC
2nd edge: CB
3rd edge: SB
4th edge: YM



Length of minimum spanning tree
$= 275 + 150 + 60 + 530$
$= 1015$ km

16. Networks, STD2 N2 2023 HSC 17

a.



b. Hobart to Darwin (with 1 stop):

$$\begin{aligned}\text{Distance} &= H \to S + S \to D \\ &= 1040 + 3150 \\ &= 4190 \text{ km}\end{aligned}$$

---

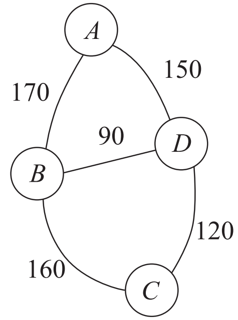
17. Networks, STD2 N2 SM-Bank 20

	A	B	C	D
A	0	4	**2**	**-**
B	4	0	**2**	**6**
C	**2**	**2**	0	3
D	**-**	6	**3**	0

---

18. Networks, STD2 N2 SM-Bank 28

a.



b. Shortest Path from A (visiting all stations)

$A–B–D–C$

$$\begin{aligned}\text{Distance} &= 170 + 90 + 120 \\ &= 380 \text{ km}\end{aligned}$$

---

19. Networks, STD2 N2 SM-Bank 37

i. ACT has 1 border (with NSW)

$\therefore$ Degree of ACT's vertex $= 1$

ii. NSW is Vertex B (it is connected to the ACT – Vertex D)

$\Rightarrow C$ is Victoria as it has degree 2

$\therefore$ Queensland is vertex A as it is connected to B and has degree 3.

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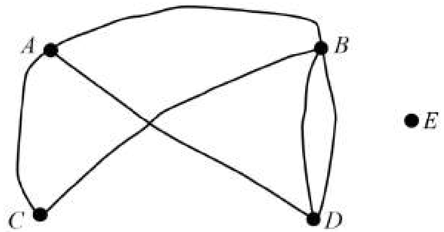
20. Networks, STD2 N2 2009 FUR2 1

a. E has no land borders with other suburbs.

b.i. & ii.



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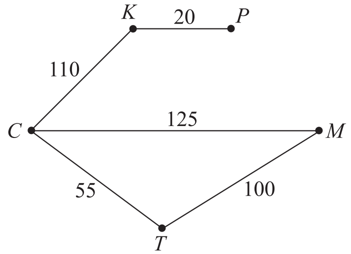
21. Networks, STD1 N1 2019 HSC 17



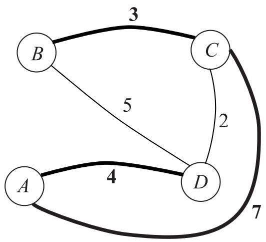
Edge weights are in minutes duration.

22. Networks, STD2 N2 SM-Bank 11

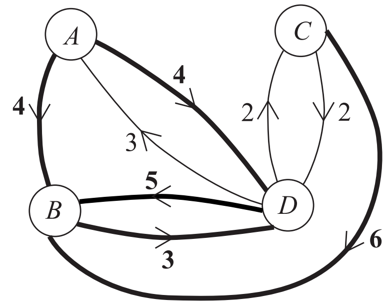
	A	*B*	*C*	*D*	*E*
A	0	10	**17**	10	10
B	**10**	0	**25**	**—**	**12**
C	17	**25**	0	—	**—**
D	**10**	—	—	0	17
E	10	**12**	**—**	**17**	0

Note the symmetry in this table across the diagonal.

23. Networks, STD2 N2 SM-Bank 21



24. Networks, STD2 N2 SM-Bank 22



25. Networks, STD2 N2 SM-Bank 23

		To			
		A	*B*	*C*	*D*
From	*A*	-	3	**0**	0
	B	2	-	**0**	**5**
	C	**0**	**4**	-	4
	D	**0**	**7**	**0**	-

26. Networks, STD2 N2 SM-Bank 40